Chapter 19 Homework Answer Sheet

Part A Multiple choice (16 points)

1-5 ACACC

6-10 DDEDB

11-16 ABEEEA

Part B Short questions (14 points)

17. (4 points)

(a) (2 Pts) $\Delta H° < 0$; $\Delta S° < 0$

(b) (2 Pts) Yes, the reaction will have $K > 1$ at low temperature.

18. (10 points)

(a) (4 Pts) $\Delta H^o = \Delta H_f^o[CH_3COOH(l)] - \Delta H_f^o[CH_3OH(l)] - \Delta H_f^o[CO(g)]$

$= –487.0 – (–238.6) – (–110.5) = –137.9$ kJ

$\Delta S^o = S°[CH_3COOH(l)] – S°[CH_3OH(l)] – S°[CO(g)]$

$= 159.8 – 126.8 – 197.9 = –164.9$ J/K $= –0.1649$ kJ/K

$\Delta G^o = \Delta H^o – T\Delta S^o = –137.9\text{ kJ} – 298\text{ K}(–0.1649\text{ kJ/K}) = –88.8$ kJ

(b) (3 Pts) $\Delta H^o = –137.9\text{ kJ} < 0$, so the reaction is exothermic. The value of K will decrease with increasing temperature, and the mole fraction of CH_3COOH will also decrease. Elevated temperatures must be used to increase the speed of the reaction.

(c) (3 Pts) $\Delta G^o = –RT\ln K$; $K = 1$, $\ln K = 0$, $\Delta G° = 0$

$\Delta G^o = \Delta H^o – T\Delta S^o$; when $\Delta G^o = 0$, $\Delta H^o = T\Delta S^o$

$–137.9\text{ kJ} = T(–0.1649\text{ kJ/K})$, $T = 836.3$ K

Thus, this reaction has an equilibrium constant equal to 1 at 836.3 K or 563.1°C.